Research Topics in Blockchain
&
Blockchain and Its Application



Lecture # 11:
Why this Consensus mechanism Works

✓ Why Consensus Algorithm works

Use case Stealing Bitcoin
Denial of Service Attack
Double Spend Attack

✓Incentive and Proof of work

✓Putting it all to geather

Usecase-1

Bitcoin’s consensus algorithm relies on cryptographic security and decentralized validation to prevent malicious activities. If an adversary, Alice, tries to manipulate the system, she faces strict cryptographic barriers.

1.Signature Verification Ensures Ownership

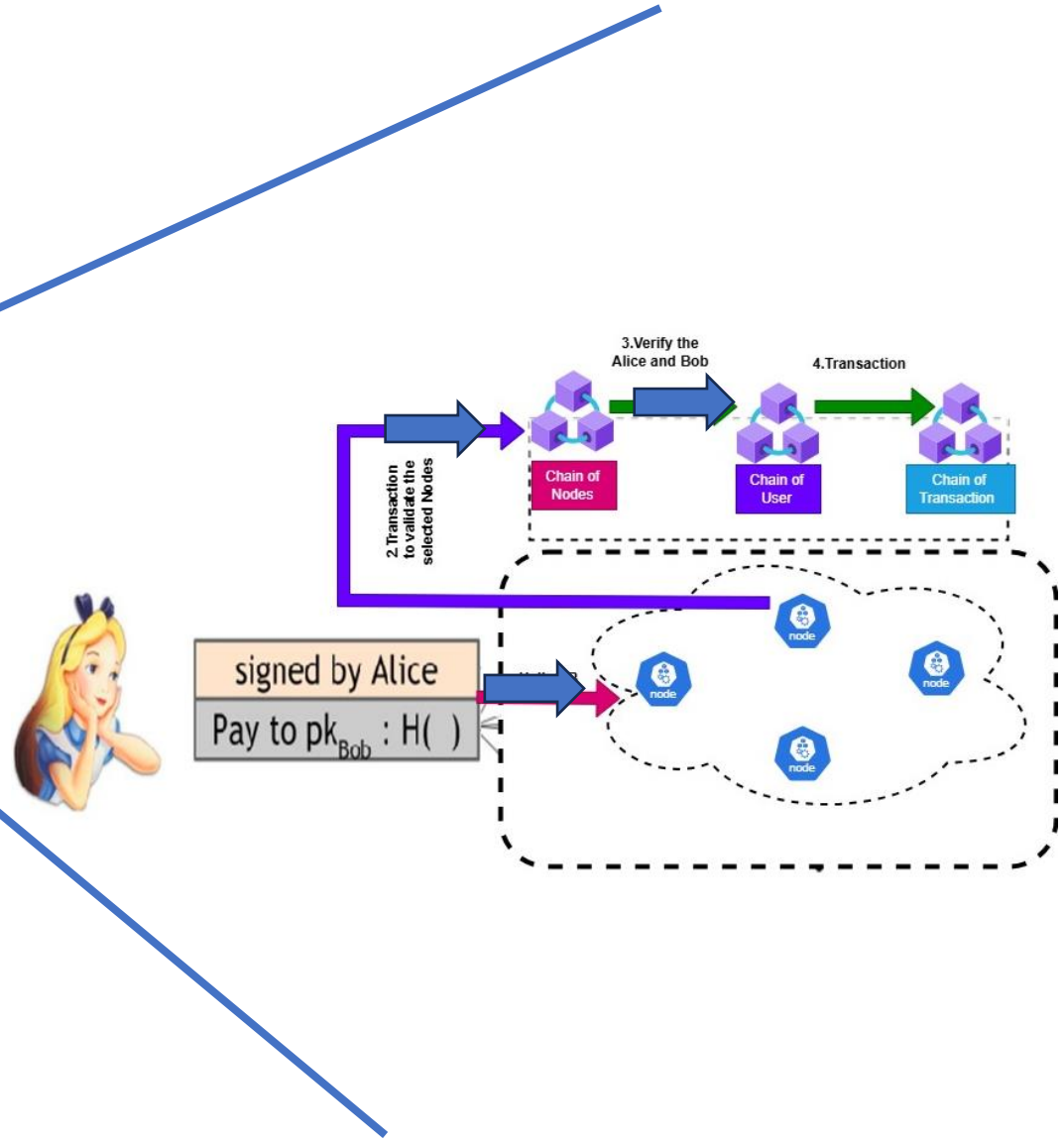
1. Every Bitcoin transaction requires a **valid digital signature** from the rightful owner.
2. Even if Alice is chosen to propose the next block, she cannot forge another user’s signature.

2.Decentralized Validation Blocks Fraud

1. All transactions must be verified by the network before inclusion in the blockchain.
2. Even if Alice attempts to include a fraudulent transaction, honest nodes will reject it.

3.Immutable Ledger Prevents Unauthorized Changes

1. Once a transaction is confirmed and added to the blockchain, it **cannot be altered or reversed** without controlling the majority of computational power (51% attack).
2. This ensures that Alice cannot retroactively change ownership records.



Mitigates Denial-of-Service Attacks

Bitcoin's consensus mechanism ensures that no single entity can permanently block transactions, even if a malicious actor like Alice attempts a **Denial-of-Service (DoS) attack** against another user, Bob.

1.Block Proposers Rotate Randomly

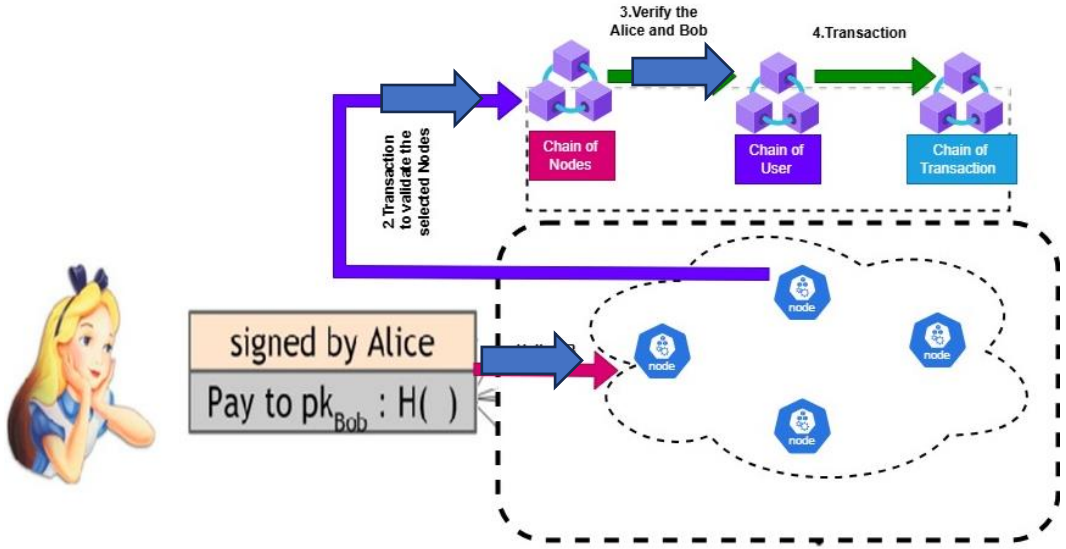
1. Alice might refuse to include Bob's transactions in her proposed block.
2. However, the **next block proposer is randomly selected**, meaning Bob's transaction will be included by another honest node.

2.Network Redundancy Prevents Censorship

1. Bitcoin's decentralized nature ensures that multiple nodes validate transactions.
2. Even if Alice ignores Bob's transaction, it will remain in the **mempool** until a fair miner includes it.

3.Transaction Persistence Ensures Inclusion

1. Transactions **do not expire** and remain valid until confirmed.
2. Eventually, Bob's transaction will be processed, making Alice's attack ineffective.



Double-Spend Attack

- **Alice's Initial Transaction:**
  - Alice buys software from Bob and sends a Bitcoin transaction to his address.
  - An honest node includes this transaction in the blockchain.
- **Bob Assumes Payment is Confirmed:**
  - Bob sees the transaction in the blockchain and allows Alice to download the software.
- **Alice's Attack (Double Spend Attempt):**
  - If Alice is selected to propose the next block, she creates a new block **ignoring** the one with her payment to Bob.
  - Instead, she includes a transaction sending the same bitcoins to another address she controls.

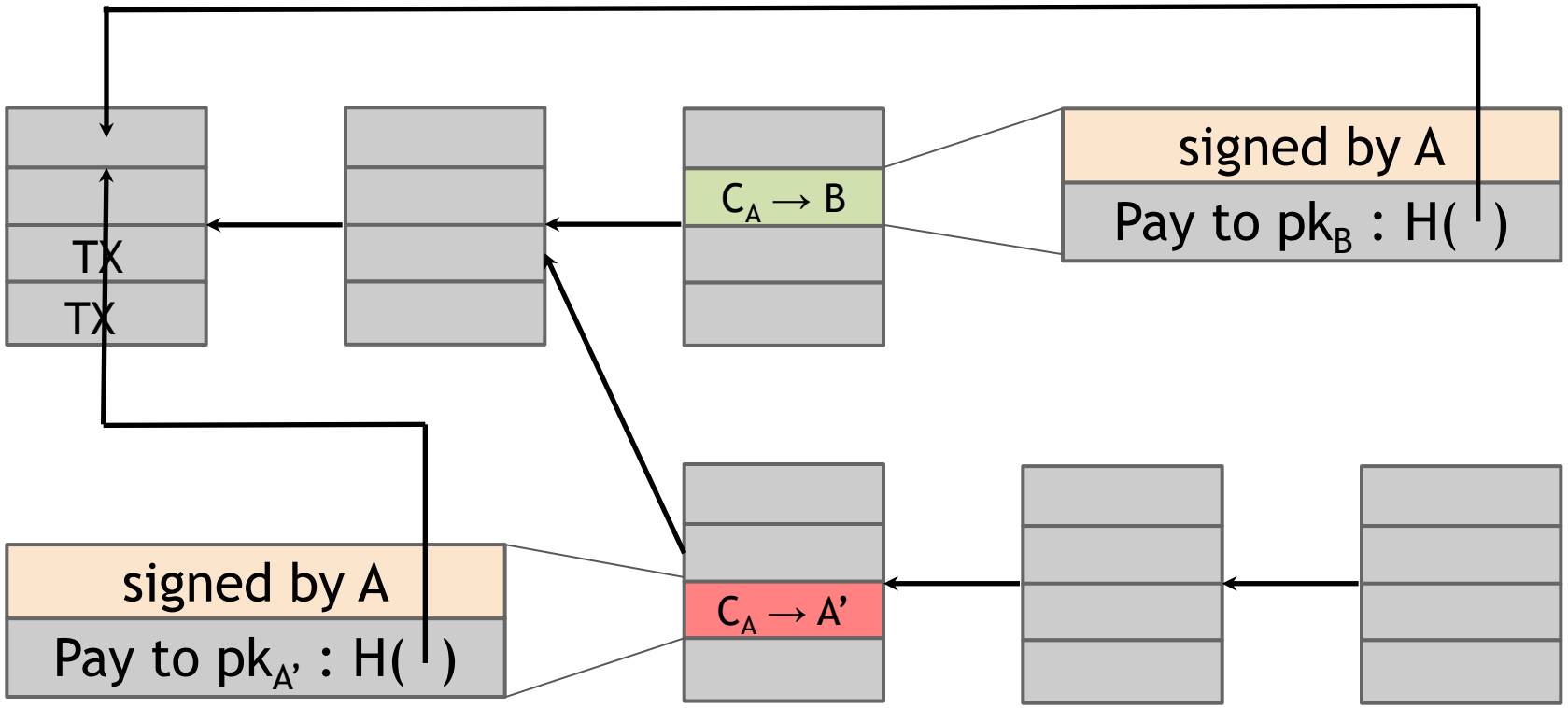
Honest Chain (longest valid chain) Block 50 → Block 51 (Txn: Alice → Bob) → Block 52

Malicious Chain (Alice's forked chain) Block 50 → Block 51' (Txn: Alice → Alice) → Block 52'

- **Conflict in the Blockchain:**
  - Two competing chains exist:
    - One where Alice pays Bob.
    - One where Alice pays herself.
  - Since both chains are the same length, the network can extend either one.
- **Outcome Depends on Network Consensus:**
  - The longest valid chain is accepted by honest nodes.
  - If honest nodes extend Bob's chain, Alice's attack fails.
  - If they extend Alice's chain, Bob never gets paid, and Alice successfully double-spends

- If the **honest chain grows longer**, the network **ignores Alice's forked chain** and follows the longest valid chain.
- If **Alice controls 51% of miners**, she can extend her fraudulent chain faster, making it the longest and causing Bob's transaction to disappear.

What can a malicious node do?



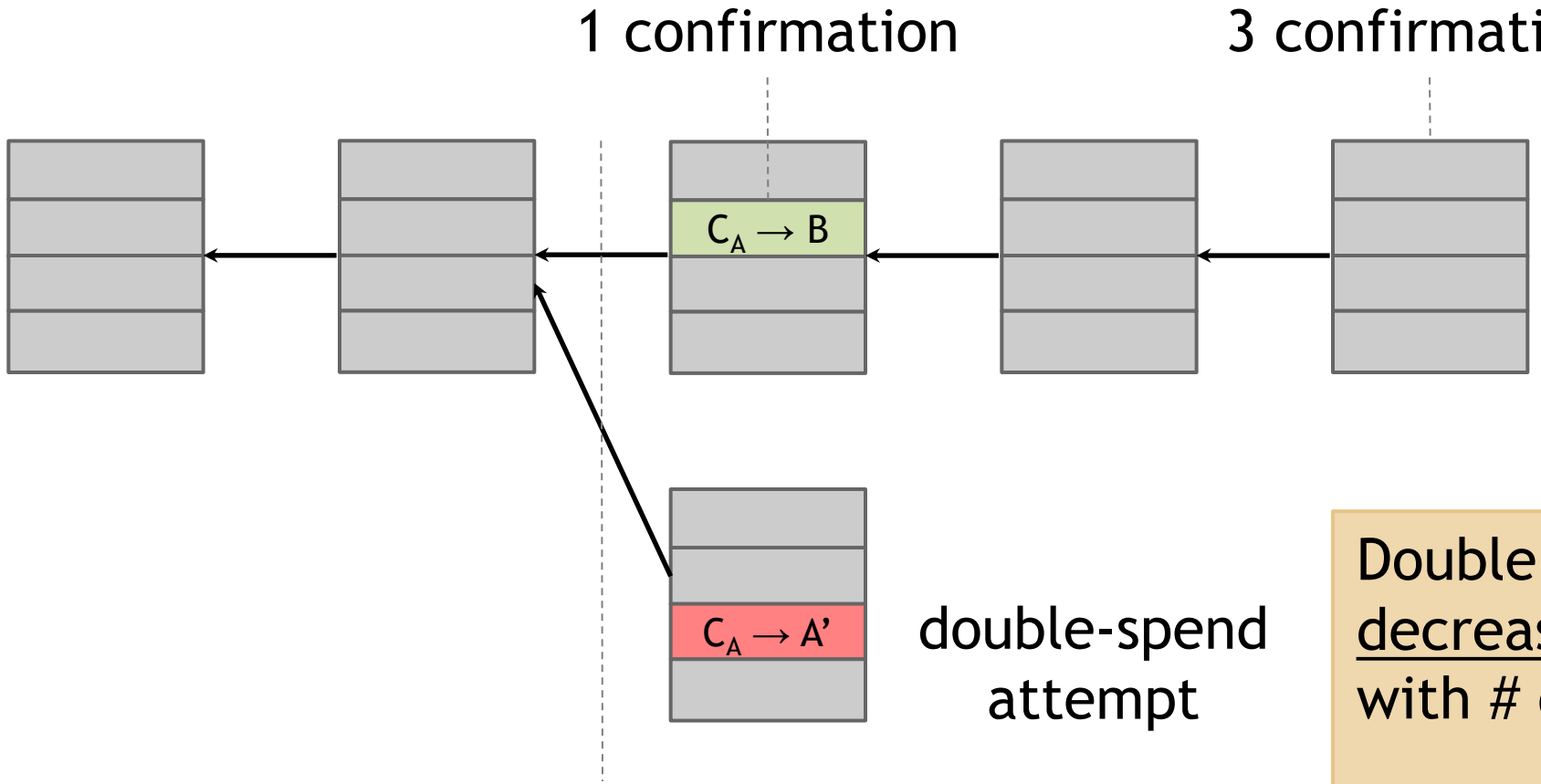
Double-spending attack

TX

Honest nodes will extend the longest valid branch

How this stop ??

From Bob the merchant's point of view



Double-spend probability <u>decreases exponentially</u> with # of confirmations

Most common heuristic:
6 confirmations

Alice Sends Bitcoin to Bob & Becomes the Miner

Step 1: Alice Creates a Transaction

- Alice wants to send 1 BTC to Bob.
- She signs the transaction with her private key and broadcasts it to the network.
- The transaction enters the mempool (waiting area for unconfirmed transactions).

Step 2: Proof-of-Work (PoW) Begins

- All miners in the network compete to solve a cryptographic puzzle.
- The miner who solves it first gets to add a new block to the blockchain.
- In this case, Alice wins the mining race and gets to propose the next block. attempt unless she controls over 51% of the mining power, which is highly unlikely.

Step 3: Alice Mines the Block

- Since Alice is the miner, she includes her own transaction (Alice → Bob, 1 BTC) in the new block.
- She solves the PoW puzzle, attaches the block to the longest valid chain, and broadcasts it.
- Other nodes in the network receive the block and verify:
  - Is the PoW valid? ✅ Are all transactions legitimate? ✅

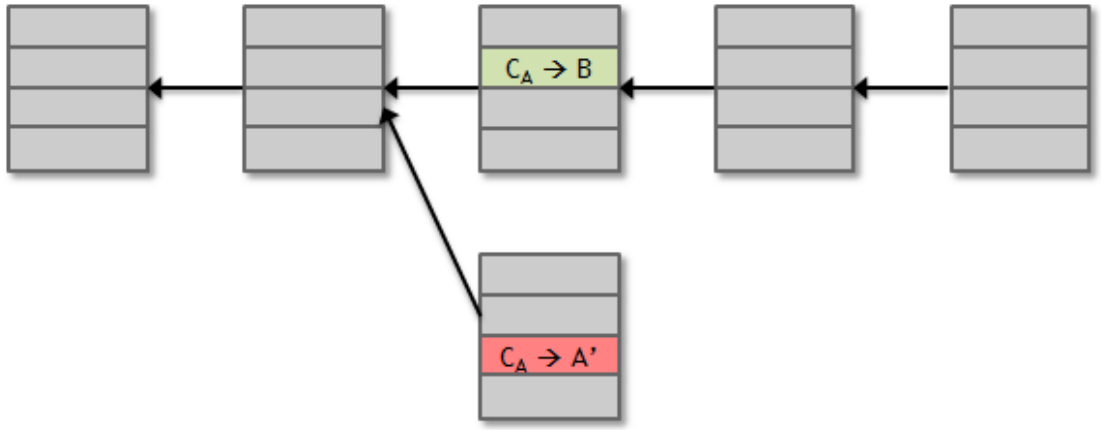
Step 4: Network Confirmation Process

- Other nodes start building on Alice's block, extending the blockchain.
- As new blocks are added, Bob waits for 6 confirmations before fully trusting the transaction.
- After 6 confirmations, Bob is sure that the transaction is final and irreversible.

Potential Attack: Alice Tries to Double Spend

- If Alice is dishonest, she could try to create another transaction sending the same 1 BTC to herself instead of Bob.
- However, for this attack to succeed, Alice must:
  - Mine a competing chain faster than the honest network.
  - Convince other miners to follow her fraudulent chain.
- If honest miners keep following the longest valid chain (which includes the transaction to Bob), Alice's double-spend attempt fails.

Recap



Protection against invalid transactions is cryptographic, but enforced by consensus

Protection against double-spending is purely by consensus

You're never 100% sure a transaction is in consensus branch. Guarantee is probabilistic

Incentives and proof of work

Discusses the incentive mechanism in Bitcoin that ensures nodes behave honestly in the consensus process.

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Nodes in the Bitcoin network do not have real-world identities, making punishment difficult.

•**Solution: Incentivizing Nodes with Bitcoin:**
•Instead of punishing dishonest nodes, Bitcoin rewards hone
•Since Bitcoin itself is a digital currency, it is used as a rewar

•**Block Reward Mechanism:**

•The miner who successfully creates a block gets a reward t
coin-creation transaction.

•**Bitcoin Halving Process:**
•Initially, the block reward was **50 BTC**, but it halves every **2**
years).
•The reward was **25 BTC** at the time of writing and continues
•This ensures a finite total supply of **21 million bitcoins**.
•The reward is assigned to an address chosen by the miner, usually their own.

•When a miner successfully mines (solves the cryptographic puzzle) and adds a new block to the blockchain, they receive a reward.
•This reward comes in the form of newly created bitcoins (or the native cryptocurrency of the blockchain) and is known as the **block reward**.
•The reward is given through a **special transaction** called the **coinbase transaction**, which is the first transaction in every block.
•The miner includes their own address in this transaction, allowing them to claim the newly minted coins.

Let's assume the following scenario where **Alice, Bob, and a Miner** are involved in transactions:

Step 1: Initial Balances (Before Any Transactions)

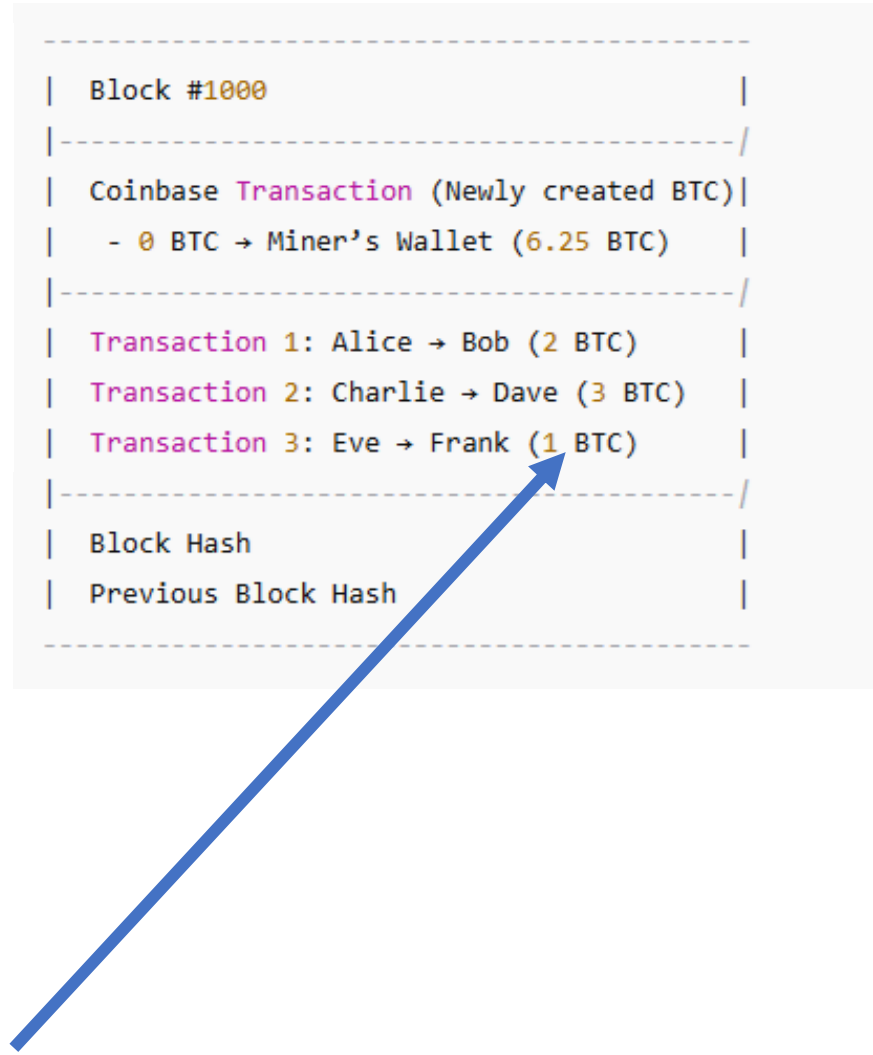
Address (Wallet)	Balance (BTC)
Alice	10 BTC
Bob	5 BTC
Miner	0 BTC

Step 2: Alice Sends 2 BTC to Bob

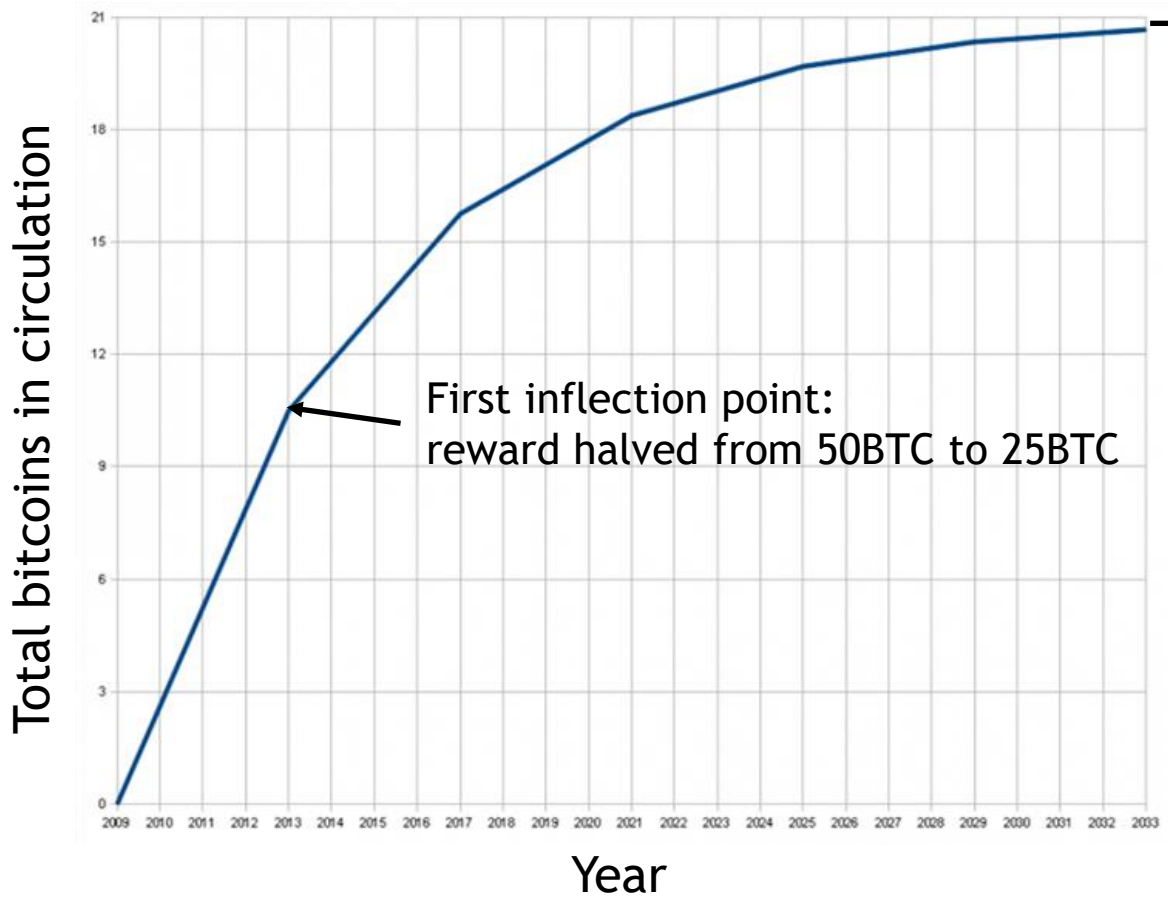
- Alice creates a transaction:
  Alice → Bob (2 BTC)
- This transaction is broadcast to the network.
- Miners compete to add this transaction into a block.

Step 3: The Miner Successfully Mines the Block

- The miner solves the cryptographic puzzle and creates a new block.
- The block contains:
  - Alice's transaction (**Alice → Bob, 2 BTC**).
  - A **coinbase transaction**, which rewards the miner with **6.25 BTC** (current Bitcoin block reward).



There’s a finite supply of bitcoins



Total supply: 21 million

Block reward is how new bitcoins are created

Runs out in 2040. No new bitcoins unless rules change

Incentive 2: transaction fees

What is a Transaction Fee in Bitcoin?

- A **transaction fee** is an additional amount that users pay to miners when they send Bitcoin (or any cryptocurrency). This fee incentivizes miners to prioritize and confirm transactions faster.

Why Do Transaction Fees Exist?

1. **Compensation for Miners** – Miners receive transaction fees as a reward for validating and adding transactions to the blockchain.
2. **Faster Confirmation** – Transactions with **higher fees** are processed more quickly since miners prioritize them.
3. **Security & Spam Prevention** – Fees prevent users from flooding the network with unnecessary transactions

Example of a Bitcoin Transaction with Fees:

Imagine **Alice** wants to send **1 Bitcoin** to **Bob**.

•The total amount she has in her Bitcoin wallet is **1.001 BTC**.

•She decides to send **1 BTC** to Bob and **0.001 BTC** as a transaction fee.

Transaction Breakdown:

- **Sender:** Alice
- **Receiver:** Bob
- **Total Sent: 1 BTC**
- **Transaction Fee: 0.001 BTC**
- **Final Amount in Bob's Wallet: 1 BTC**
- **Miners Receive: 0.001 BTC as a reward**

Remaining problems

1. How to pick a random node?

1. How to avoid a free-for-all due to rewards?

1. How to prevent Sybil attacks?

Proof of work

To approximate selecting a random node:
select nodes in proportion to a resource
that no one can monopolize (we hope)

- In proportion to computing power: proof-of-work
- In proportion to ownership: proof-of-stake

Equivalent views of proof of work

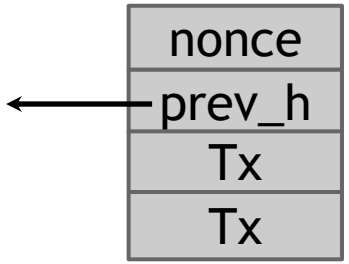
1. Select nodes in proportion to computing power

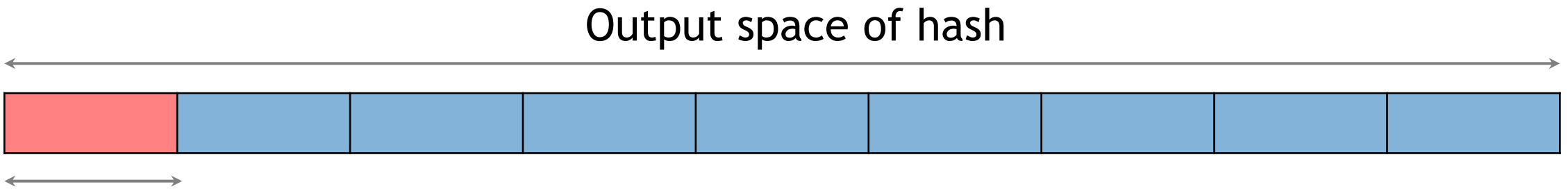
1. Let nodes compete for right to create block

1. Make it moderately hard to create new identities

Hash puzzles

To create block, find nonce s.t.
H(nonce ‖ prev_hash ‖ tx ‖ … ‖ tx) is very small





If hash function is secure:
only way to succeed is to try enough nonces until you get lucky

PoW property 1: difficult to compute

As of Aug 2014: about 10^{20} hashes/block

Only some nodes bother to compete — miners

PoW property 2: parameterizable cost

Nodes automatically re-calculate the target every two weeks

Goal: average time between blocks = 10 minutes

Prob (Alice wins next block) =
fraction of global hash power she controls

Every 2016 blocks (~two weeks), the Bitcoin network automatically recalculates this target to maintain an average block time of **10 minutes**.

- If blocks are being mined too quickly (less than 10 minutes on average), the difficulty **increases**.
- If blocks are being mined too slowly (more than 10 minutes on average), the difficulty **decreases**.

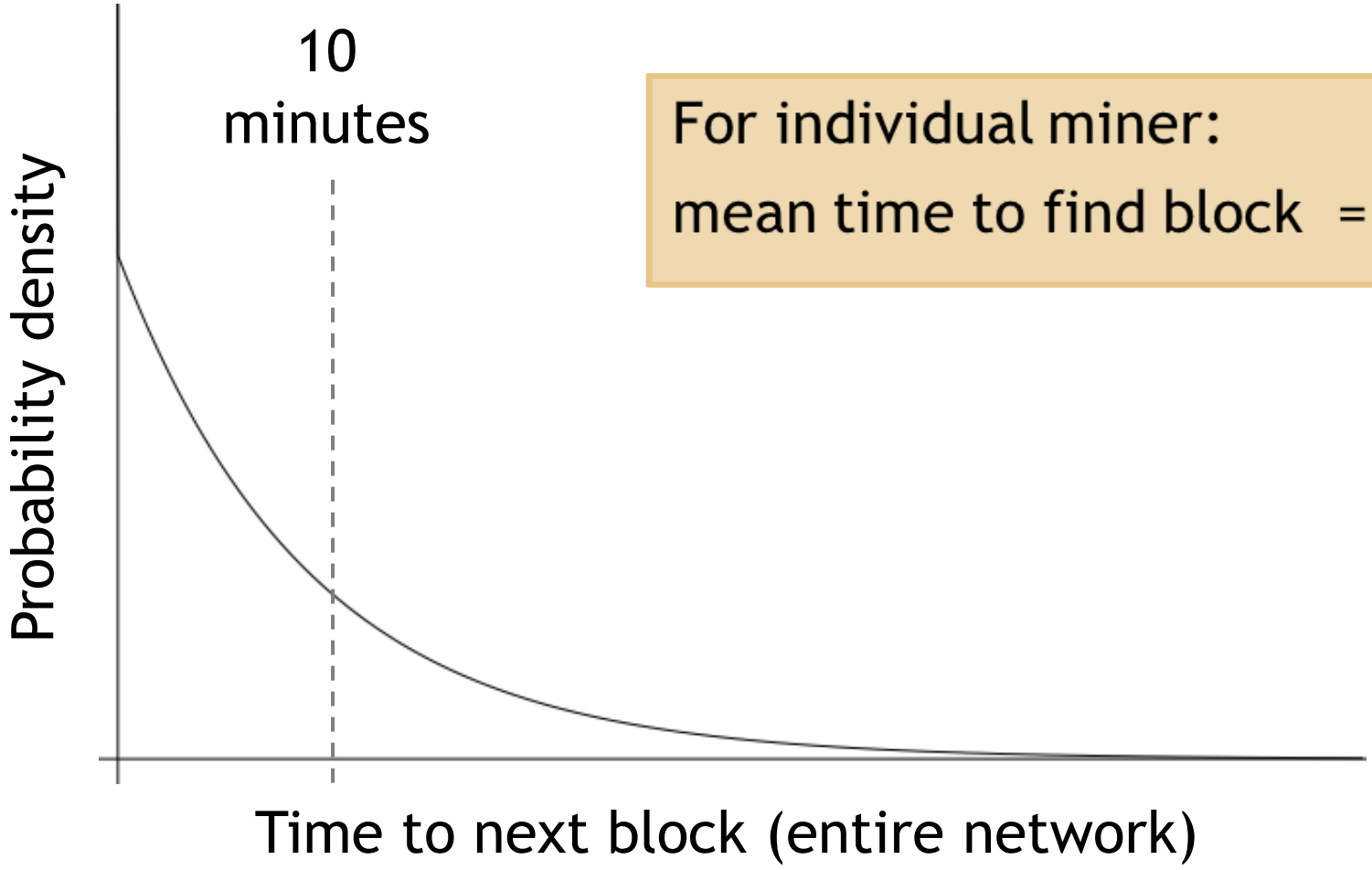
Why is this important?

- It keeps the Bitcoin network stable and predictable, **preventing blocks from being mined too fast or too slow**.
- It ensures fairness and prevents a sudden increase in mining

Key security assumption

Attacks infeasible if majority of miners weighted by hash power follow the protocol

Solving hash puzzles is probabilistic



For individual miner:

$$\text{mean time to find block} = \frac{10 \text{ minutes}}{\text{fraction of hash power}}$$

PoW property 3: trivial to verify

Nonce must be published as part of block

Other miners simply verify that
$H(\text{nonce} \| \text{prev_hash} \| \text{tx} \| \ldots \| \text{tx}) < \text{target}$

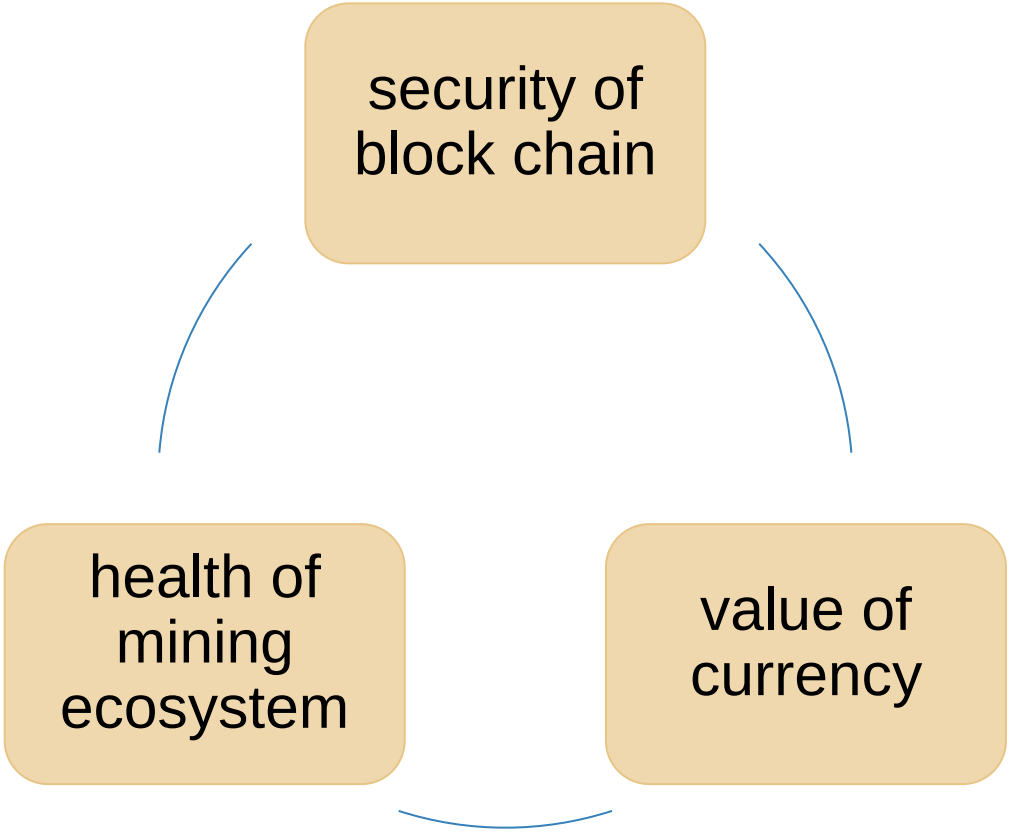
Mining economics

If mining reward (block reward + Tx fees)	>	hardware + electricity cost	→	Profit

Complications:

- fixed vs. variable costs
- reward depends on global hash rate

Bitcoin is bootstrapped



What can a “51% attacker” do?

Steal coins from existing address? ✗

Suppress some transactions?

- From the block chain ✓
- From the P2P network ✗

Change the block reward? ✗

Destroy confidence in Bitcoin? ✓✓

Remaining questions

How do we get from consensus to currency?

What else can we do with consensus?